



Regulatory Complaint Investigation and CAPA Report

Confidential - AI Draft

Document No.	DP-M4-20260708-ALARMGOE
Classification	Confidential - AI Draft
Generated	2026-07-08T09:30:48Z
Device	DocPlus+ Pulse Oximeter (DEV-PULSE-OX)
Firmware	v3.4
Framework	ISO 13485
Prepared By	DocPlus+ M4 Evidence and Documentation Layer

This document was generated from live M1 graph facts and M2 agent outputs. Every summarized claim is tied to captured agent state, evidence citations, or audit findings.

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This output follows a formal regulatory-document pattern: controlled metadata first, then complaint facts, investigation logic, objective evidence, risk decision, CAPA plan, review gates, and approval placeholders.

1. Document Profile and Controlled Metadata

Document Title	DocPlus+ Regulatory Complaint Investigation and CAPA Report
Document Number	DP-M4-20260708-ALARMGOE
Revision	A - AI Generated Draft
Complaint ID	CPL-20260708-C257
CAPA ID	CAPA-20260708-DA21
RTM ID	RTM-DP-C257DA21
Classification	Confidential - AI Draft for quality and regulatory review
Generated Timestamp	2026-07-08T09:30:48Z
System of Record	DocPlus+ M4 Evidence and Documentation Layer
Device	DocPlus+ Pulse Oximeter (DEV-PULSE-OX)
Current Firmware	v3.4
Affected Component	alarm_manager
Regulatory Framework	ISO 13485

Data Quality and Source Provenance

This report references 46 graph fact(s): 12 controlled, 22 extracted, 6 inferred, 5 synthetic, 1 internal.

Uncontrolled graph facts appear in this report and are visibly tagged. [DEMO DATA - NOT CONTROLLED EVIDENCE] means the item cannot support CAPA closure without controlled verification. [INFERRED DATA - REVIEW REQUIRED] means the item was inferred and requires reviewer confirmation before final use.

Source Type	Referenced Fact Count
controlled	12
extracted	22
inferred	6
synthetic	5
internal	1

Digital-Twin Prototype Disclaimer

Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Document Purpose

This report documents the complaint intake, AI-assisted investigation record, retrieved objective evidence, risk decision, CAPA recommendations, audit-style gaps, and corrective action controls. It is intended as a controlled draft that a Quality or Regulatory reviewer can verify, approve, or return for correction.

2. Regulatory Format Controls

ALCOA+ Data Integrity Mapping

Principle	DocPlus+ Implementation
Attributable	Prepared by DocPlus+ M4; generated timestamp and device scope are recorded.
Legible	A4 PDF with selectable text, structured headings, repeatable tables, and page numbers.
Contemporaneous	Generated at the time of complaint-report request from live agent state.
Original	Marked as AI Draft until Quality and Regulatory reviewers approve.
Accurate	Risk, CAPA, evidence, audit, and trace sections are populated from structured state.
Complete	Includes intake, investigation, evidence, risk, CAPA, audit, trace, and approval fields.
Consistent	Uses the same DocPlus+ controlled layout for all complaint-report exports.
Enduring	Saved as a PDF artifact under the application output directory.
Available	Downloadable through the M4 document endpoint and traceable by document number.

Applicable Regulatory Basis

Framework	Relevance to This Report
ISO 13485	Complaint handling, CAPA discipline, documented information, traceable quality records.
21 CFR Part 820	Design controls, complaint files, CAPA, and objective evidence expectations.
EU MDR	Technical documentation, post-market surveillance, vigilance, and risk-benefit continuity.
21 CFR Part 11	Electronic record controls, auditability, attribution, and durable retrieval.

3. Executive Summary

Severity High	Risk Critical / RPN 20	Reportable True
Evidence Confidence medium (score 0.95)	Evidence Class Mix Controlled current verification: 10, Firmware traceability signal - informational, not controlled verification evidence: 1	Evidence Items 11
Audit Findings 1	Trace Decay Alerts 2	

Complaint: alarm goes on even when the level of SpO2 is normal

Agent conclusion: Complaint states: "alarm goes on even when the level of SpO2 is normal" for DEV-PULSE-OX (severity High). The affected component aligns with alarm_manager. Controlled verification records for firmware v3.4 show passing results for multiple alarm requirements focused on missing-alarm scenarios (RUN-TC-ALRM-051/052/053/054), which do not directly address false-positive behavior at normal SpO2. Prior, closed complaints (CPL-100-051, CPL-100-052) involved alarms not triggering and were resolved via field correction with identified software causes; these are adjacent but not the same failure mode. Based on supplied hypotheses and evidence, root cause is not confirmed; most plausible are an inadvertent alarm_manager software bug or incorrect threshold configuration. Additional device-specific data and targeted reproduction are required.

Risk Evidence Confidence Basis

Weighted evidence-class score 0.95 from 11 item(s): Controlled current verification=10, Firmware traceability signal - informational, not controlled verification evidence=1. CAPA-blocking evidence-class item count: 1. Rule: Risk evidence confidence is the average of per-item evidence-class weights (controlled_verification=1.00, historical_controlled=0.70, simulated=0.45, candidate=0.25, no_evidence=0.00). The final label is capped below high when any CAPA-closure-blocking evidence class is present, because those items cannot support final closure until controlled verification is attached.

Confidence drivers: RUN-TC-ALRM-054-V3-4: controlled_verification, RUN-TC-ALRM-052-V3-4: controlled_verification, RUN-TC-ALRM-051-V3-4: controlled_verification, RUN-TC-ALRM-053-V3-4: controlled_verification, RUN-TC-ALRM-056-V3-4: controlled_verification, RUN-TC-ALRM-058-V3-4: controlled_verification, RUN-TC-ALRM-055-V3-4: controlled_verification, RUN-TC-ALRM-057-V3-4: contro...

Quality Review Decision

The complaint should remain open until objective evidence is reconciled, CAPA ownership is assigned, and any major or critical audit findings affecting the complaint scope are remediated or justified.

4. Complaint Investigation Record

Field	Generated Value	Regulatory Basis
Complaint ID	CPL-20260708-C257	ISO 13485 8.2.2
Date Received	2026-07-08T09:30:48Z	21 CFR Part 11
Device UDI	UDI-DI-08717648-236474	EU MDR Article 27 / 21 CFR 830
Model / Version	DOCPLUS+-PULSE-OXIMETER / v3.4	IEC 62304 Section 8
Complaint Source	Post-market complaint intake	ISO 13485 8.2.2
Complaint Narrative	alarm goes on even when the level of SpO2 is normal	21 CFR 820 complaint file
Patient Impact	Potential delayed awareness of a safety or status condition.	EU MDR Article 87
MDR Reportable?	Yes	EU MDR Article 87 / 21 CFR 803
CAPA Initiated?	Yes - CAPA-20260708-DA21	21 CFR 820.100
Affected Component	alarm_manager	ISO 13485 8.5.2
Closure Status	Pending Engineering Approval - Simulated Evidence Sufficient: Formal Engineering review must approve the simulated/minor-gap findings and decide whether additional physical verification is required.	ISO 13485 8.2.2

Reportability Decision Logic

Condition	DocPlus+ Assessment	Action
Death or serious injury	Not stated in complaint text.	Monitor for escalation during intake and investigation.
Potential harm or near miss	Potential delayed awareness of a safety or status condition.	Treat as reportability review required.
Device relationship	Complaint maps to Alarm Notification, Measurement and Sensor Performance.	Keep CAPA open until engineering evidence confirms or rules out relationship.
Malfunction/use issue	Reported condition indicates possible device malfunction or use-related failure.	Verify recurrence risk and effectiveness of controls.

Related Prior Complaints

Complaint ID	Similarity	Summary	Source
CPL-100-051	3	High SpO2 alarm does not trigger. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-052	3	Low SpO2 alarm does not trigger. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CMP-005	2	Alarm sounded late during simulated desaturation. [DEMO DATA - NOT CONTROLLED EVIDENCE]	synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE]
CMP-012	2	MAUDE-style adverse event record linked to pulse oximeter use. [DEMO DATA - NOT CONTROLLED EVIDENCE]	synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE]
CPL-100-039	2	Artificial nails prevent detection. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]

5. Requirements Traceability Matrix

Req ID	Requirement	Scope / Source	Relevance	Evidence	Test Case	Status / Result
REQ-ALR M-051	High SpO2 alarm does not trigger.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 7.	verified_current - Controlled current verification	TC-ALRM-051	current and usable - Pass
REQ-ALR M-052	Low SpO2 alarm does not trigger.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 7.	verified_current - Controlled current verification	TC-ALRM-052	current and usable - Pass
REQ-ALR M-053	High pulse rate alarm missing.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 6.	verified_current - Controlled current verification	TC-ALRM-053	current and usable - Pass
REQ-ALR M-054	Low pulse rate alarm missing.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 6.	verified_current - Controlled current verification	TC-ALRM-054	current and usable - Pass
REQ-ALR M-055	Alarm sounds continuously without reason.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 6.	verified_current - Controlled current verification	TC-ALRM-055	current and usable - Pass
REQ-ALR M-056	Alarm volume cannot be adjusted.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 6.	verified_current - Controlled current verification	TC-ALRM-056	current and usable - Pass
REQ-ALR M-057	Alarm remains muted permanently.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 6.	verified_current - Controlled current verification	TC-ALRM-057	current and usable - Pass
REQ-ALR M-058	Alarm delay exceeds specification.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 6.	verified_current - Controlled current verification	TC-ALRM-058	current and usable - Pass
REQ-ALR M-059	Alarm icon does not appear.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 6.	verified_current - Controlled current verification	TC-ALRM-059	current and usable - Pass
REQ-ALR M-060	Alarm continues after condition is cleared.	strong - RTM_M aster.xlsx	Matched complaint terms against alarm_manager; relevance score 6.	verified_current - Controlled current verification	TC-ALRM-060	current and usable - Pass
REQ-SPO 2-003	Alarm behavior shall notify the user when saturation drops below configured thresholds. [DEMO DATA - NOT CONTROLLED EVIDENCE]	strong - M1 requirement scope [DEMO DATA - NOT CONTROLLED EVIDENCE]	Matched complaint terms against Design Control; relevance score 6.	missing - Candidate evidence - requires Quality review [DEMO DATA - NOT CONTROLLED EVIDENCE]	TEST-003, TEST-023	usable with caution - Pass, Needs Review
REQ-MEA S-039	Artificial nails prevent detection.	weak - RTM_M aster.xlsx	Matched complaint terms against measurement_core; relevance score 2.	verified_current - Controlled current verification	TC-MEAS-039	current and usable - Pass

Trace Decay Risk Statement

Rows marked STALE are not filler rows: they are generated from the live Trace Decay result and indicate verification links that require review before the device can claim current-firmware evidence.

6. Design History File Index

#	Element	Document ID	Generated Unique Content	Status
1	Planning	DDP-DP-C257DA21	Design plan for complaint-impact review	Open
2	Design Input	URS-DP-C257DA21	Complaint-linked requirement review for alarm goes on even when the level of SpO2 is normal	Draft
3	Design Output	SDD-DP-C257DA21	Alarm Notification, Measurement and Sensor Performance design output references mapped	Draft
4	Design Review	DRM-DP-C257DA21	Cross-functional review required	Pending
5	Verification	VPR-DP-C257DA21	Objective evidence required for Alarm Notification, Measurement and Sensor Performance on v3.4	Open
6	Validation	VAL-DP-C257DA21	Use-condition validation required for alarm goes on even when the level of SpO2 is normal	Open
7	Design Transfer	DTR-DP-C257DA21	Release readiness blocked until evidence reconciled	Pending
8	Design Changes	DCR-DP-C257DA21	Change/service impact assessment linked to Alarm Notification, Measurement and Sensor Performance	Open
9	DHF Index	DHFI-DP-C257DA21	DocPlus+ index for DocPlus+ Pulse Oximeter	Generated

Validation Gap Check

The validation section remains open because the complaint describes a user-reported condition involving Alarm Notification, Measurement and Sensor Performance. DocPlus+ therefore does not mark the DHF complete until validation, verification, or justified non-reproducibility evidence is linked.

7. EU MDR Technical Documentation - Annex II / III

Annex Ref	Section	DocPlus+ Generated Content
Annex II Section 1	Device Description	UDI UDI-DI-08717648-236474 scoped to complaint involving Alarm Notification, Measurement and Sensor Performance.
Annex II Section 3	Design and Manufacturing Information	Design/change impact for Alarm Notification, Measurement and Sensor Performance is carried into design output review.
Annex II Section 4	GSPR Compliance	Reported condition maps to safety/performance review: alarm goes on even when the level of SpO2 is normal.
Annex II Section 5	Benefit-Risk Analysis	Critical risk complaint requires residual-risk review before closure.
Annex II Section 6.1	Verification and Validation	Objective verification evidence must be reconciled against the complaint scope.
Annex III Section 1.3	PMS / PSUR Input	Complaint is retained as a post-market signal for trend monitoring.
Annex III Section 1.5	Vigilance Reporting	Reportability remains open until human regulatory review confirms decision.

Team NB Structure Check

The package preserves a cohesive structure with section titles, metadata, and cross-references so that information is locatable by a reviewer rather than buried in a generic AI narrative.

Record Boundary

The report is generated only from live application state: M1 graph context, M2 agent outputs, M4 evidence retrieval, AuditShadow findings, and Trace Decay alerts. It does not certify regulatory closure until the signature block is completed by authorized reviewers.

8. Complaint Investigation Workflow and Execution Trace

8.1 Structured Complaint Intake Detail

Device ID	DEV-PULSE-OX
Affected Component	alarm_manager (COMP-ALARM-MANAGER)
Severity	High
Timeline	not specified
Symptom Codes	SYMPTOM_ALARM, SYMPTOM_GOES, SYMPTOM_EVEN, SYMPTOM_WHEN, SYMPTOM_LEVEL
Match Terms	alarm, spo2
Similar Prior Complaints	5

Complaint Narrative

alarm goes on even when the level of SpO2 is normal

Related Prior Complaints

Complaint ID	Similarity	Summary	Source
CPL-100-051	3	High SpO2 alarm does not trigger. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-052	3	Low SpO2 alarm does not trigger. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CMP-005	2	Alarm sounded late during simulated desaturation. [DEMO DATA - NOT CONTROLLED EVIDENCE]	synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE]
CMP-012	2	MAUDE-style adverse event record linked to pulse oximeter use. [DEMO DATA - NOT CONTROLLED EVIDENCE]	synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE]
CPL-100-039	2	Artificial nails prevent detection. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]

8.2 Investigation Chronology

Time	Agent	Action	Status	Detail
2026-07-08T09:27:44.061587Z	langgraph_supervisor	route to complaint_intake	info	Recorded
2026-07-08T09:27:44.061587Z	complaint_intake	extract structured complaint	started	Recorded
2026-07-08T09:30:14.935033Z	complaint_intake	extract structured complaint	completed	Elapsed 150873.5 ms
2026-07-08T09:30:14.935033Z	langgraph_supervisor	handoff check: similar incidents preserved for root cause	info	Structured event metadata recorded
2026-07-08T09:30:14.955237Z	langgraph_supervisor	route to root_cause	info	Recorded
2026-07-08T09:30:14.955237Z	root_cause	generate complaint-specific failure hypotheses	started	Recorded
2026-07-08T09:30:19.940138Z	root_cause	generate complaint-specific failure hypotheses	completed	Elapsed 4984.9 ms
2026-07-08T09:30:19.985982Z	langgraph_supervisor	route to firmware_traceability_ripple_check	info	Recorded
2026-07-08T09:30:19.986501Z	firmware_traceability_ripple_check	run firmware traceability ripple checks	started	Recorded
2026-07-08T09:30:19.993113Z	firmware_traceability_ripple_check	hypothesis capability check	info	Structured event metadata recorded
2026-07-08T09:30:22.067852Z	firmware_traceability_ripple_check	hypothesis capability check	info	Structured event metadata recorded
2026-07-08T09:30:22.068853Z	firmware_traceability_ripple_check	hypothesis capability check	info	Structured event metadata recorded
2026-07-08T09:30:22.068853Z	firmware_traceability_ripple_check	run firmware traceability ripple checks	completed	Elapsed 2082.8 ms
2026-07-08T09:30:22.146605Z	langgraph_supervisor	route to evidence	info	Recorded
2026-07-08T09:30:22.146605Z	evidence	retrieve and validate evidence	started	Recorded
2026-07-08T09:30:22.568235Z	evidence	component-scoped evidence retrieval	info	Structured event metadata recorded
2026-07-08T09:30:22.569247Z	evidence	retrieve and validate evidence	completed	Elapsed 422.2 ms
2026-07-08T09:30:22.631380Z	langgraph_supervisor	route to risk	info	Recorded

Chronology Interpretation

The sequence confirms the complaint passed through intake, root-cause hypothesis generation, firmware traceability ripple checking, evidence retrieval, risk scoring, CAPA drafting, and final reasoning. This preserves a reviewer-readable process trail for the AI investigation rather than presenting the CAPA as an unsupported narrative.

Trace AI - Pipeline Observability

Trace AI is operational engineering telemetry for this report generation. It records pipeline execution, LLM metadata, and retrieval/tool metadata only. It is not clinical evidence, not regulatory objective evidence, and not the same as Trace Decay or the Firmware Traceability Ripple Check.

Total wall-clock time	440419.11 ms
Agent steps	10
LLM calls	4
Tool/retrieval calls	25
Tokens	23932 total (16595 prompt / 7337 completion)
Longest agent	complaint_intake (150873.5 ms)

Agent Step Trace

Step ID	Agent	Action	Status	Elapsed	Sub-events
step-e1d7461253	cybersecurity	scan SBOM components against NVD	success	99897.6 ms	LLM 0; tools 3
step-df3538d913	complaint_intake	extract structured complaint	success	150873.5 ms	LLM 1; tools 4
step-14d811bf1d	audit_shadow	run adversarial audit	success	138580.64 ms	LLM 0; tools 5
step-4b73817f6f	openai_reasoning	final audit reasoning	success	17514.08 ms	LLM 1; tools 0
step-f80720e3e7	root_cause	generate complaint-specific failure hypotheses	success	4984.94 ms	LLM 1; tools 2
step-c7b0ddd2cd	firmware_traceability_ripple_...	run firmware traceability ripple checks	success	2082.69 ms	LLM 0; tools 2
step-f56d8f5353	evidence	retrieve and validate evidence	success	422.17 ms	LLM 0; tools 3
step-272c73aca	risk	calculate ISO 14971 risk	success	186.31 ms	LLM 0; tools 2
step-6711c6ad9f	capa	draft CAPA with citations	success	2892.53 ms	LLM 0; tools 4
step-913435ed02	openai_reasoning	final complaint reasoning	success	22984.65 ms	LLM 1; tools 0

LLM Call Metadata

Call ID	Agent	Task	Model	Status	Tokens	Latency
llm-1ae89e7a08	complaint_intake	symptom_classification	gpt-5	success	2063 (1358/705)	6188.28 ms
llm-ffda768495	root_cause	root_cause_hypotheses	gpt-4o-mini	success	5595 (4839/756)	4527.1 ms
llm-120dd30822	openai_reasoning	audit_final_reasoning	gpt-5	success	7665 (5527/2138)	17512.31 ms
llm-1170a4ec05	openai_reasoning	complaint_final_reasoning	gpt-5	success	8609 (4871/3738)	22980.49 ms

Tool and Retrieval Call Metadata

Call ID	Agent	Tool	Query / Target	Results	Latency
tool-44710853e3	cybersecurity	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		38.12 ms
tool-9127aa281d	cybersecurity	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		49.33 ms
tool-c6ff6b8090	complaint_intake	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		78.03 ms
tool-2d3bfb286	audit_shadow	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		114.35 ms
tool-c9354e3c12	complaint_intake	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		173.34 ms
tool-22cd7a1654	audit_shadow	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		179.02 ms
tool-ae30498bde	cybersecurity	graph_tools.get_device_context	device_id=DEV-PULSE-OX	15	99818.6 ms
tool-c0a4979f57	audit_shadow	graph_tools.get_device_context	device_id=DEV-PULSE-OX	15	101268.44 ms
tool-c7865b72dd	complaint_intake	graph_tools.get_device_context	device_id=DEV-PULSE-OX	15	101412.75 ms
tool-58f5478f0c	audit_shadow	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		41.77 ms
tool-f5b18377f2	audit_shadow	graph_tools.get_readiness	device_id=DEV-PULSE-OX		37052.12 ms
tool-4d36b7520b	complaint_intake	graph_tools.find_similar_complaints	term=alarm goes on even when the level of SpO2 is normal SYMPTOM_ALARM...	5	45.93 ms
tool-a322c6c604	root_cause	graph_tools.get_graph_neighborhood	node_id=COMP-ALARM-MANAGER	417	75.33 ms
tool-a698339acb	root_cause	graph_tools.get_graph_neighborhood	node_id=COMP-ALARM-MANAGER	14	378.77 ms
tool-5204dd60da	firmware_traceability_rip...	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		96.05 ms
tool-835f2ab568	firmware_traceability_rip...	graph_tools.run_ripple	device_id=DEV-PULSE-OX; new_version=v3.5; changed_components=["COMP-ALA...	10	2059.65 ms
tool-68351a0fb3	evidence	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		94.86 ms
tool-29116f354c	evidence	graph_tools.get_component_scoped_ev...	device_id=DEV-PULSE-OX; component_id=COMP-ALARM-MANAGER; component_name...	12	349.3 ms
tool-35cf6c4f72	evidence	vector_tools.retrieve_documents_debug	device_id=DEV-PULSE-OX; query=alarm goes on even when the level of SpO2...	0	70.66 ms
tool-70768ec362	risk	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		64.64 ms

5 additional tool/retrieval call(s) are retained in structured Trace AI data.

9. Root Cause Analysis

ID	Hypothesis	Probability	Affected Area	Key Why Chain
HYP-001	alarm_manager: Incorrect Alarm Threshold Configuration [INFERRED DATA - REVIEW REQUIRED]	33%	COMP-ALARM-MANAGER	Alarm thresholds are intended to alert based on specific SpO2 levels. If the thresholds are incorrectly set or coded, the alarm may trigger unnecessarily. [INFERRED DATA - REVIEW REQUIRED]
HYP-002	alarm_manager: Inadvertent Software Bug [INFERRED DATA - REVIEW REQUIRED]	39%	COMP-ALARM-MANAGER	Software bugs can lead to logic errors in alarm processing algorithms. If sensor data is misinterpreted, it could cause the alarm to activate incorrectly. [INFERRED DATA - REVIEW REQUIRED]
HYP-003	alarm_manager: Sensor Calibration Error [INFERRED DATA - REVIEW REQUIRED]	28%	COMP-ALARM-MANAGER	Calibration errors can lead to incorrect readings from the SpO2 sensor, which could be interpreted as fluctuations or either high/low saturation. If sensors are miscalibrated, they may falsely signal alarms. [INFERRED DATA - REVIEW REQUIRED]

HYP-001 Evidence Balance

Evidence For	Prior complaint (CPL-100-051) indicated issues with SpO2 alarms not triggering properly suggests configuration errors support this hypothesis. [INFERRED DATA - REVIEW REQUIRED]
Evidence Against / Gaps	No evidence currently indicating exact threshold configurations or test results to confirm this issue. [INFERRED DATA - REVIEW REQUIRED]
Similar Incident Cross-Check	CPL-100-051: Consistent pattern with alarm thresholds causing issue for SpO2; closure indicates a similar issue with alarm behavior. [INFERRED DATA - REVIEW REQUIRED]
Probability Rationale	Moderate evidence from similar incidents implies potential threshold issues, but lacks direct evidence of current configuration problems. [INFERRED DATA - REVIEW REQUIRED]

HYP-002 Evidence Balance

Evidence For	Complaint report states the observed failure condition: alarm goes on even when the level of SpO2 is normal [INFERRED DATA - REVIEW REQUIRED]
Evidence Against / Gaps	No evidence currently showing that the alarm manager's software is malfunctioning; would require debugging findings. No direct evidence, but previous complaints (CPL-100-052) indicate software-related issues affecting alarm functionality. [INFERRED DATA - REVIEW REQUIRED]
Similar Incident Cross-Check	CPL-100-052: Inconsistent pattern seen where alarms do not trigger, which might relate to software state; unclear linkage. [INFERRED DATA - REVIEW REQUIRED]
Probability Rationale	Higher likelihood due to software dependencies and vague reports of similar incidents affecting alarm responses, but lacking hard evidence. [INFERRED DATA - REVIEW REQUIRED]

HYP-003 Evidence Balance

Evidence For	Previous complaints (CPL-100-039) involve disruptions related to sensor performance, providing context for possible sensor misreads. [INFERRED DATA - REVIEW REQUIRED]
Evidence Against / Gaps	No direct evidence of calibration records or current calibration status available to confirm this as the root cause. [INFERRED DATA - REVIEW REQUIRED]
Similar Incident Cross-Check	CMP-005: Some alignment with feedback on alarm timing which could tangentially support calibration issues but lacks specificity. Insufficient evidence across similar incidents to directly link sensor calibration to current alarm behavior. [INFERRED DATA - REVIEW REQUIRED]
Probability Rationale	Moderate balance of indications from similar cases relative to sensor performance but not directly linked to alarm triggers. [INFERRED DATA - REVIEW REQUIRED]

Root Cause Rationale

1. The reported symptom ("alarm goes on even when the level of SpO2 is normal") maps to alarm_manager and aligns with the structured complaint fields and terms (alarm, spo2).
2. Controlled verification evidence indicates REQ-ALRM-051/052/053/054 passed on firmware v3.4 (RUN-TC-ALRM-051/052/053/054). These focus on conditions where alarms fail to trigger (and related behaviors), not on spurious alarms at normal SpO2; therefore, the supplied passing results neither confirm nor refute the current false-positive allegation.
3. Similar incidents: CPL-100-051 (root cause: High SpO2 alarm comparator threshold issue) and CPL-100-052 (root cause: Low SpO2 alarm inhibit timer issue) were closed with field correction initiated. While they demonstrate prior software-related alarm defects, their failure modes were "alarm does not trigger," differing from the present "alarm triggers when SpO2 is normal," so they are suggestive but not determinative.
4. Use/environmental factor example CPL-100-039 ("Artificial nails prevent detection") shows user/environment effects can impact readings; however, it addressed detection failure rather than false alarms at normal SpO2 and provides limited direct support for this case.
5. Hypothesis assessment using supplied bases: HYP-002 (Inadvertent Software Bug, base_probability 0.39) is most plausible given the symptom and history of software-related alarm behaviors; HYP-001 (Incorrect Alarm Threshold Configuration, base_probability 0.33) remains credible given prior threshold-comparator issues; HYP-003 (Sensor Calibration Error, base_probability 0.28) has the weakest direct support from the provided set.
6. Critical gaps: firmware version, serial/lot, event date/timeline, alarm settings at time of event, and objective device logs are missing; without these, root cause cannot be confirmed from the supplied evidence set.

Primary Root Cause Position

For this generated draft, the leading position is that the complaint is associated with Alarm Notification, Measurement and Sensor Performance and the evidence/control set tied to the reported condition. The position remains provisional until engineering review confirms or rules out the suspected cause with objective evidence.

10. Evidence and Provenance

Source	Evidence Class	Support	Conf.	Excerpt	Citation
REQ-ALRM-054 / TC-ALRM-054 / RUN-TC-ALRM-054-V3-4	Controlled current verification	Yes	96%	REQ-ALRM-054: Low pulse rate alarm missing. Test TC-ALRM-054: Verification test TC-ALRM-054; result=Pass; firmware=v3.4. Evidence RUN-TC-ALRM-054-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4 Ver...	[Source: REQ-ALRM-054/TC-ALRM-054, Confidence: 0.96]
REQ-ALRM-052 / TC-ALRM-052 / RUN-TC-ALRM-052-V3-4	Controlled current verification	Yes	96%	REQ-ALRM-052: Low SpO2 alarm does not trigger. Test TC-ALRM-052: Verification test TC-ALRM-052; result=Pass; firmware=v3.4. Evidence RUN-TC-ALRM-052-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4...	[Source: REQ-ALRM-052/TC-ALRM-052, Confidence: 0.96]
REQ-ALRM-051 / TC-ALRM-051 / RUN-TC-ALRM-051-V3-4	Controlled current verification	Yes	96%	REQ-ALRM-051: High SpO2 alarm does not trigger. Test TC-ALRM-051: Verification test TC-ALRM-051; result=Pass; firmware=v3.4. Evidence RUN-TC-ALRM-051-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4...	[Source: REQ-ALRM-051/TC-ALRM-051, Confidence: 0.96]
REQ-ALRM-053 / TC-ALRM-053 / RUN-TC-ALRM-053-V3-4	Controlled current verification	Yes	96%	REQ-ALRM-053: High pulse rate alarm missing. Test TC-ALRM-053: Verification test TC-ALRM-053; result=Pass; firmware=v3.4. Evidence RUN-TC-ALRM-053-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4 Ve...	[Source: REQ-ALRM-053/TC-ALRM-053, Confidence: 0.96]
REQ-ALRM-056 / TC-ALRM-056 / RUN-TC-ALRM-056-V3-4	Controlled current verification	Yes	96%	REQ-ALRM-056: Alarm volume cannot be adjusted. Test TC-ALRM-056: Verification test TC-ALRM-056; result=Pass; firmware=v3.4. Evidence RUN-TC-ALRM-056-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4...	[Source: REQ-ALRM-056/TC-ALRM-056, Confidence: 0.96]

Source	Evidence Class	Support	Conf.	Excerpt	Citation
REQ-ALRM-058 / TC-ALRM-058 / RUN-TC-ALRM-058-V3-4	Controlled current verification	Yes	96%	REQ-ALRM-058: Alarm delay exceeds specification. Test TC-ALRM-058: Verification test TC-ALRM-058; result=Pass; firmware=v3.4. Evidence RUN-TC-ALRM-058-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3....	[Source: REQ-ALRM-058/TC-ALRM-058, Confidence: 0.96]

10.1 Firmware Traceability Ripple Check

This section reports a firmware-traceability check: for each root-cause hypothesis, it examines whether linked verification test records match the complaint or candidate firmware version, or are stale/mismatched. This is a graph-based traceability signal, not a behavioral model or physical test of the device. It identifies where verification evidence may be out of date, not what the device would actually do under the reported failure condition.

Hypothesis	Analysis Type	Result	Interpretation	Evidence Class
HYP-002	Firmware traceability ripple check	fail	HYP-002 evaluates alarm_manager: Inadvertent Software Bug. The firmware traceability ripple check found 10 linked verification test record(s) that do not match candidate firmware v3.5; this flags stale or mismatched verification coverage for this hypothesis...	Firmware traceability signal - informational, not controlled verification evidence
HYP-001	Not applicable - measurement_calibration_or_physiology	not_run	HYP-001 evaluates alarm_manager: Incorrect Alarm Threshold Configuration. The firmware traceability ripple check was not run because this failure mode maps to 'measurement_calibration_or_physiology', outside firmware-to-test traceability freshness review. T...	Firmware traceability signal - informational, not controlled verification evidence
HYP-003	Not applicable - measurement_calibration_or_physiology	not_run	HYP-003 evaluates alarm_manager: Sensor Calibration Error. The firmware traceability ripple check was not run because this failure mode maps to 'measurement_calibration_or_physiology', outside firmware-to-test traceability freshness review. This hypothesis...	Firmware traceability signal - informational, not controlled verification evidence

11. CAPA Report

Product	DocPlus+ Pulse Oximeter
Software Version	v3.4
CAPA Status	Pending Engineering Approval - Simulated Evidence Sufficient
Priority	High
Risk Classification	Major - potential impact to Alarm Notification, Measurement and Sensor Performance
Evidence Confidence	medium (score 0.95)
Evidence Class Mix	Controlled current verification: 10, Firmware traceability signal - informational, not controlled verification evidence: 1

Risk Evidence Confidence Basis

Weighted evidence-class score 0.95 from 11 item(s): Controlled current verification=10, Firmware traceability signal - informational, not controlled verification evidence=1. CAPA-blocking evidence-class item count: 1. Rule: Risk evidence confidence is the average of per-item evidence-class weights (controlled_verification=1.00, historical_controlled=0.70, simulated=0.45, candidate=0.25, no_evidence=0.00). The final label is capped below high when any CAPA-closure-blocking evidence class is present, because those items cannot support final closure until controlled verification is attached.

Engineering Approval Gate

This CAPA is supported by 10 controlled verification record(s); 1 simulated result(s) and 0 minor evidence-class gap(s) remain below the materiality threshold (0%). Formal Engineering review must approve the simulated/minor-gap findings and decide whether additional physical verification is required. Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Problem Statement

A high priority complaint was opened for DocPlus+ Pulse Oximeter. The user reported: alarm goes on even when the level of SpO2 is normal. In practical terms, the concern is that a user may not receive a safety or status notification at the expected time. The complaint record confirms the reported condition; the exact technical cause and final correction remain under investigation until objective evidence is reviewed.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Complaint Summary

Reported condition: alarm goes on even when the level of SpO2 is normal. Affected area: Alarm Notification, Measurement and Sensor Performance. Reported timing or trigger: During active use or monitoring. Observed symptom signals: inconsistent or inaccurate measurement readings. No patient or user injury is stated in the complaint text; the risk review must still confirm potential safety impact.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Immediate Containment Actions

Preserve the complaint evidence for Alarm Notification, Measurement and Sensor Performance, including returned-unit information, event logs, service records, photos, user statements, configuration, lot or serial traceability, and software or hardware version where applicable. Screen recent complaints for similar symptoms and issue interim support, service, or use instructions if the risk review shows potential user or patient impact.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Investigation

Relevant device requirements, verification records, risk controls, and open quality actions were reviewed for the complaint scope. The investigation focused on the reported condition and the affected area: Alarm Notification, Measurement and Sensor Performance. For this complaint the scoped area is Alarm Notification, Measurement and Sensor Performance, with emphasis on alarm or notification timing, event logs, and risk-control behavior. Engineering and Quality reviewed the available design expectations, test records, risk controls, event evidence, and related quality actions. 2 verification records need Quality review. Some verification evidence still needs Quality review before CAPA closure. Risk evidence basis: Weighted evidence-class score 0.95 from 11 item(s): Controlled current verification=10, Firmware traceability signal - informational, not controlled verification evidence=1. CAPA-blocking evidence-class item count: 1.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Root Cause Analysis

The current root-cause hypothesis is: The alarm manager may have an incorrect configuration of the SpO2 thresholds that is causing it to trigger alarms even when SpO2 levels are normal. This is a probable cause, not a final confirmed cause, because the CAPA record still needs returned-unit review, complaint evidence, trend review, and approved verification evidence showing whether the suspected cause is truly responsible.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Why CAPA Remains Open

The CAPA is not blocked for lack of evidence. The scoped evidence chain has controlled verification plus digital-twin or firmware-traceability simulation support, so the remaining gate is formal Engineering approval. Engineering must review the simulated findings against notification timing and user-visible response, approve or reject their applicability, and record whether physical/controlled verification is still required. Risk evidence confidence is medium (score 0.95); Medium uncertainty: 10 of 11 evidence item(s) are current controlled verification, but 1 item(s) remain historical, simulated, candidate, or no-evidence class and cannot support CAPA closure. This CAPA is supported by 10 controlled verification record(s); 1 simulated result(s) and 0 minor evidence-class gap(s) remain below the materiality threshold (0%). It remains pending formal Engineering approval before closure; simulated results support the investigation but do not substitute for physical or controlled verification. Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Risk Assessment

The main hazard is that a user may not receive a safety or status notification at the expected time. The current risk level is Critical and the complaint should remain in active risk review under ISO 13485 until patient impact, recurrence potential, and residual risk acceptability are documented. Risk evidence confidence is medium (score 0.95); Medium uncertainty: 10 of 11 evidence item(s) are current controlled verification, but 1 item(s) remain historical, simulated, candidate, or no-evidence class and cannot support CAPA closure.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Corrective Actions

Reproduce or simulate the reported condition for Alarm Notification, Measurement and Sensor Performance under the reported use conditions. Software/System Engineering should review alarm or notification timing, event logs, and risk-control behavior, identify the confirmed failure mode, correct the issue or document why it cannot be reproduced. Engineering must review and approve the simulated findings before closure; if Engineering rejects the simulation or deems it insufficient, open targeted physical or controlled verification before final approval.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Preventive Actions

Update the design, service, or release checklist so future changes affecting Alarm Notification, Measurement and Sensor Performance include complaint-specific verification for inconsistent or inaccurate measurement readings. Add regression or trend monitoring where appropriate, and require evidence review before Quality approves a similar change or field disposition.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Verification of Effectiveness

Effectiveness is supported by controlled verification records plus simulated analysis of notification timing and user-visible response. Before closure, an Engineering reviewer must approve the digital-twin or firmware-traceability simulation result, document why it is sufficient for this complaint, or require additional physical verification.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

CAPA Effectiveness Criteria

The CAPA is effective when the behavior for notification timing and user-visible response is supported by current controlled records and the digital-twin or firmware-traceability simulation is formally approved by Engineering. Closure requires documented Engineering sign-off; additional physical verification is required only if Engineering does not accept the simulated support.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Lessons Learned

Future complaint handling for Alarm Notification, Measurement and Sensor Performance should connect user-reported symptoms, objective evidence, risk controls, and verification records early. Similar changes or service decisions should include test coverage for the reported condition and review of whether existing controls remain effective.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

CAPA Closure

Closure decision: pending Engineering approval. This CAPA is supported by 10 controlled verification record(s); 1 simulated result(s) and 0 minor evidence-class gap(s) remain below the materiality threshold (0%). Formal Engineering review must approve the simulated/minor-gap findings and decide whether additional physical verification is required. Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Technical evidence identifiers, requirement IDs, audit findings, and source citations are retained in the evidence and audit appendices. They are not repeated here so the CAPA remains readable for Quality, Engineering, Regulatory, and non-technical reviewers.

11.1 CAPA Action Plan

CAPA Record	CAPA-20260708-DA21
Complaint Reference	CPL-20260708-C257
CAPA Type	Corrective and preventive action
Current Status	Pending Engineering Approval - Simulated Evidence Sufficient
Engineering Approval Gate	This CAPA is supported by 10 controlled verification record(s); 1 simulated result(s) and 0 minor evidence-class gap(s) remain below the materiality threshold (0%). Formal Engineering review must approve the simulated/minor-gap findings and decide whether additional physical verification is required. Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Action Plan

Purpose	Owner	Required Action	How Closure Is Proven
Immediate measures	Quality / Support	Preserve the complaint evidence for Alarm Notification, Measurement and Sensor Performance, including returned-unit information, event logs, service records, photos, user statements, configuration, lot or serial traceability, and software or hardware version where applicable. Screen recent complaints for similar symptoms and issue interim support, service, or use instructions if the risk review shows potential user or patient impact.	Evidence is preserved and simulated findings are available for Engineering review.
Fix the issue	Software/System Engineering	Reproduce or simulate the reported condition for Alarm Notification, Measurement and Sensor Performance under the reported use conditions. Software/System Engineering should review alarm or notification timing, event logs, and risk-control behavior, identify the confirmed failure mode, correct the issue or document why it cannot be reproduced. Engineering must review and approve the simulated findings before closure; if Engineering rejects the simulation or deems it insufficient, open targeted physical or controlled verification before final approval.	Engineering approves the digital-twin/simulated finding or records that physical verification is required.
Prevent recurrence	Engineering / Quality	Update the design, service, or release checklist so future changes affecting Alarm Notification, Measurement and Sensor Performance include complaint-specific verification for inconsistent or inaccurate measurement readings. Add regression or trend monitoring where appropriate, and require evidence review before Quality approves a similar change or field disposition.	Future changes include Engineering review of any simulated evidence before approval.
Check effectiveness	Quality Assurance	Effectiveness is supported by controlled verification records plus simulated analysis of notification timing and user-visible response. Before closure, an Engineering reviewer must approve the digital-twin or firmware-traceability simulation result, document why it is sufficient for this complaint, or require additional physical verification.	Closure is proven by Engineering sign-off on the simulated support plus Quality/Regulatory approval.

Acceptance Criteria for Closure

Closure Gate	Required Evidence
Reported condition	The reported condition is supported by controlled records and simulated investigation findings.
Affected function	Notification timing and user-visible response is covered by the scoped controlled and simulated evidence chain.
Engineering review	Engineering formally approves the digital-twin or firmware-traceability simulation, or requires physical verification.
Prototype disclaimer	Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.
Approval	Engineering, Quality, and Regulatory reviewers approve the completed CAPA record before closure.

12. Cybersecurity / SBOM Narrative

This section is device-level cybersecurity surveillance. It is not complaint evidence and it does not change the Risk Agent score automatically. It records what the Cybersecurity Agent found by reading the SBOM and querying the National Vulnerability Database for the CPE candidates in the workbook.

SBOM components	35
Components queried	25
NVD findings	7
Query date	2026-07-04
Source type	extracted
Cache status	cache_hit

Executive Severity Rollup

Critical	High	Medium	Low	None
0	2	5	0	0

VEX-style exploitability status is shown per CVE. The current automated SBOM/NVD match does not include device-specific exploitability determinations, so findings default to Under Investigation until Engineering documents whether the vulnerable code path is affected, fixed, not affected, or not reachable in this device configuration.

SBOM Component Coverage

Component	Version	Supplier	CPE Candidate	NVD Result
Bootloader	1.8.0	Device OEM / Secure Boot Team	cpe:2.3:o:device_oem:pulseox_bootloader:1.8.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Embedded Firmware	2.4.3	Device OEM / Embedded Software Team	cpe:2.3:o:device_oem:pulseox_firmware:2.4.3:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Operating System / RTOS	10.5.1	Amazon Web Services, Inc.	cpe:2.3:o:amazon:freertos:10.5.1:****:*	open_cves_found: 1 CVE record(s) returned by NVD for this CPE.
Device Drivers	1.6.2	Device OEM / Driver Integration Team	cpe:2.3:a:device_oem:pulseox_device_drivers:1.6.2:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Signal Processing Software	3.2.1	Device OEM / Clinical Algorithm Supplier	cpe:2.3:a:device_oem:pulseox_signal_processing:3.2.1:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Measurement Control Logic	2.1.0	Device OEM / Embedded Software Team	cpe:2.3:a:device_oem:pulseox_measurement_control:2.1.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Alarm Management Module	2.0.5	Device OEM / Safety Software Team	cpe:2.3:a:device_oem:pulseox_alarm_manager:2.0.5:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
User Interface Module	1.7.4	Device OEM / UI Software Team	cpe:2.3:a:device_oem:pulseox_ui_module:1.7.4:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Settings Configuration Manager	1.9.1	Device OEM / Embedded Software Team	cpe:2.3:a:device_oem:pulseox_settings_manager:1.9.1:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Data Storage Manager	1.5.0	Device OEM / Data Platform Team	cpe:2.3:a:device_oem:pulseox_data_storage_manager:1.5.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Patient Record Manager	1.3.2	Device OEM / Data Platform Team	cpe:2.3:a:device_oem:pulseox_patient_record_manager:1.3.2:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.

Component	Version	Supplier	CPE Candidate	NVD Result
Communication Stack	2.2.0	Device OEM / Connectivity Team	cpe:2.3:a:device_oem:pulseox_communication_stack:2.2.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Bluetooth Stack	5.3.0	Nordic Semiconductor ASA	cpe:2.3:a:nordicsemi:bluetooth_low_energy_stack:5.3.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Wi-Fi Stack	2.10	w1.fi project / Device OEM Integration	cpe:2.3:a:w1.fi:wpa_supplicant:2.10:****:	open_cves_found: 1 CVE record(s) returned by NVD for this CPE.
TCP/IP Stack	2.1.3	lwIP project	cpe:2.3:a:lwip_project:lwip:2.1.3:****:***	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
USB Interface Software	1.4.1	Device OEM / Connectivity Team	cpe:2.3:a:device_oem:pulseox_usb_interface:1.4.1:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
File System	3.8.1	littlefs project / Device OEM Integration	cpe:2.3:a:littlefs_project:littlefs:3.8.1:****:***	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Logging Module	1.2.8	Device OEM / Platform Software Team	cpe:2.3:a:device_oem:pulseox_logging_module:1.2.8:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Error Handling Module	1.1.6	Device OEM / Platform Software Team	cpe:2.3:a:device_oem:pulseox_error_handling:1.1.6:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Time Management Module	1.0.9	Device OEM / Platform Software Team	cpe:2.3:a:device_oem:pulseox_time_manager:1.0.9:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Self-Test Module	1.3.5	Device OEM / Test Software Team	cpe:2.3:a:device_oem:pulseox_self_test:1.3.5:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Firmware Update Manager	2.0.2	Device OEM / Release Engineering Team	cpe:2.3:a:device_oem:pulseox_update_manager:2.0.2:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Authentication Module	1.2.0	Device OEM / Security Software Team	cpe:2.3:a:device_oem:pulseox_authentication_module:1.2.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.

Component	Version	Supplier	CPE Candidate	NVD Result
Authorization Module	1.2.0	Device OEM / Security Software Team	cpe:2.3:a:device_oem:pulseox_authorization_module:1.2.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Cryptographic Library	3.0.13	OpenSSL Software Foundation	cpe:2.3:a:openssl:openssl:3.0.13:****:*	open_cves_found: 5 CVE record(s) returned by NVD for this CPE.

NVD CVE Findings - Structured Vulnerability Records

Vulnerability Record - CVE-2024-28115

CVE ID / NVD Reference	CVE-2024-28115 https://nvd.nist.gov/vuln/detail/CVE-2024-28115
CVSS Score / Severity / Vector	HIGH / CVSS 8.8 / v3.1 Vector: CVSS:3.1/ AV:L/ AC:L/ PR:L/ UI:N/ S:C/ C:H/ I:H/ A:H
CWE ID(s)	CWE-284: Improper Access Control, NVD-CWE-Other: Other weakness classification
Affected Component + Version	Operating System / RTOS 10.5.1
Vulnerability Description	NVD classifies the CVE under CWE-284, NVD-CWE-Other; review the public description for deployment-specific impact.
Exploitability / Exposure Context	Local or physically mediated exploitation should be assessed against service, update, or peripheral workflows. SBOM exposure: No direct exposure unless network stacks or external interfaces run as RTOS tasks.
Recommended Controls	review vendor fix status, confirm exploit preconditions, and document whether compensating controls cover the affected interface. SBOM controls: Minimal RTOS configuration, stack overflow checks, watchdog, task priority review, patch monitoring.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2024-28115

Vulnerability Record - CVE-2023-52160

CVE ID / NVD Reference	CVE-2023-52160 https://nvd.nist.gov/vuln/detail/CVE-2023-52160
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.5 / v3.1 Vector: CVSS:3.1/ AV:N/ AC:L/ PR:N/ UI:R/ S:U/ C:H/ I:N/ A:N
CWE ID(s)	CWE-287: Improper Authentication
Affected Component + Version	Wi-Fi Stack 2.10
Vulnerability Description	NVD describes an authentication or access-control bypass condition in the affected component.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Wi-Fi radio, WPA2/WPA3 negotiation, outbound cloud network.
Recommended Controls	update the TLS library, enforce strict certificate validation, and regression-test invalid certificate cases. SBOM controls: WPA2/WPA3, credential encryption, no ad-hoc mode, network allow-list, update monitoring.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2023-52160

Vulnerability Record - CVE-2009-1390

CVE ID / NVD Reference	CVE-2009-1390 https://nvd.nist.gov/vuln/detail/CVE-2009-1390
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.8 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P
CWE ID(s)	CWE-287: Improper Authentication
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes incomplete TLS certificate-chain validation, where a connection may be accepted because one certificate in an otherwise untrusted chain is trusted.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	verify full certificate-chain validation, require trusted root and hostname match, and regression-test rejected-chain cases. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-1390

Vulnerability Record - CVE-2009-3765

CVE ID / NVD Reference	CVE-2009-3765 https://nvd.nist.gov/vuln/detail/CVE-2009-3765
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.8 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P
CWE ID(s)	CWE-310: Cryptographic Issues
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes certificate host-name validation that mishandles embedded NUL characters in the certificate Common Name, enabling a crafted certificate-name mismatch.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	reject embedded NUL characters in certificate names, prefer patched TLS hostname-validation routines, and test crafted certificate names. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-3765

Vulnerability Record - CVE-2009-3766

CVE ID / NVD Reference	CVE-2009-3766 https://nvd.nist.gov/vuln/detail/CVE-2009-3766
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.8 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P
CWE ID(s)	CWE-310: Cryptographic Issues
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes a certificate or TLS validation weakness in the affected secure-communication path.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	update the TLS library, enforce strict certificate validation, and regression-test invalid certificate cases. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-3766

Vulnerability Record - CVE-2009-3767

CVE ID / NVD Reference	CVE-2009-3767 https://nvd.nist.gov/vuln/detail/CVE-2009-3767
CVSS Score / Severity / Vector	MEDIUM / CVSS 4.3 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:N/ I:P/ A:N
CWE ID(s)	CWE-295: Improper Certificate Validation
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes certificate host-name validation that mishandles embedded NUL characters in the certificate Common Name, enabling a crafted certificate-name mismatch.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	reject embedded NUL characters in certificate names, prefer patched TLS hostname-validation routines, and test crafted certificate names. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-3767

Vulnerability Record - CVE-2019-0190

CVE ID / NVD Reference	CVE-2019-0190 https://nvd.nist.gov/vuln/detail/CVE-2019-0190
CVSS Score / Severity / Vector	HIGH / CVSS 7.5 / v3.1 Vector: CVSS:3.1/ AV:N/ AC:L/ PR:N/ UI:N/ S:U/ C:N/ I:N/ A:H
CWE ID(s)	NVD-CWE-noinfo: No CWE information
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes a denial-of-service condition that may let an attacker disrupt availability of the affected software path.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	patch affected code paths, constrain exposed interfaces, and confirm watchdog/recovery behavior for availability loss. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2019-0190

13. Electronic Records - 21 CFR Part 11

Requirement	DocPlus+ Implementation	Part 11
System Validation	DocPlus+ generation path is deterministic and testable through FastAPI route and PDF render checks.	11.10(a)
Audit Trail	Generation record links CPL-20260708-C257, CAPA-20260708-DA21, timestamp, and agent trace events.	11.10(e)
Access Controls	AI output is marked draft and requires QE/RA approval before controlled use.	11.10(d)
Electronic Signatures	Signature block captures reviewer, approver, date, and meaning of signature.	11.50
Record Integrity	PDF is generated as a stable artifact with a deterministic document number and retrievable filename.	11.10(c)
Human-Readable Copy	Selectable-text PDF is available through the document download endpoint.	11.10(b)
Retention	Artifact is retained under output/pdf and can be indexed by M5 or a QMS repository.	11.10(c)

Generation Log Summary

Generation Hash	C257DA21
Input Fingerprint	C257DA21EF26C8EC
AI Draft Status	Pending human quality and regulatory review
Low Confidence Handling	Evidence below 70% must be reviewed before release; current generated evidence remains visible with confidence values.

14. AuditShadow and Trace Decay

AuditShadow findings

Finding(s)	Requirement(s)	Risk	Observation	Remediation
FINDING-020	REQ-SPO2-003	Major	M1 graph has no VERIFIED_BY test evidence linked for this requirement. M4 may have retrieved candidate evidence, but it is not controlled objective verification evidence until Quality links it to the requirement, test record, source, and version/configuration metadata. [DEMO DATA - NOT CONTROLLED EVIDENCE]	Link or create verification test evidence with current version/configuration metadata, then rerun AuditShadow. [DEMO DATA - NOT CONTROLLED EVIDENCE]

Trace Decay alerts

Requirement(s)	Test(s)	Tested FW	Required FW	Reason
REQ-SPO2-003	TEST-003, TEST-023	v3.4	v3.4	Test is synthetic/demo backfill and cannot be treated as controlled verification evidence. [DEMO DATA - NOT CONTROLLED EVIDENCE]

14.1 Audit Data Interpretation and Corrective Action

AuditShadow and Trace Decay use the live M1 traceability graph as the controlled source for verification status. When the report flags missing evidence, it means the requirement does not have a formal VERIFIED_BY test link in that graph. It does not mean M4 found no supporting material. In this run, M4 retrieved 11 candidate evidence item(s), but candidate snippets remain draft support until a Quality reviewer links them to a requirement, test record, firmware version, and controlled source identifier.

Signal	What It Means	Source	Corrective Action
AuditShadow graph evidence gap	1 requirement finding(s) lack a VERIFIED_BY test link in M1.	M1 readiness score and traceability matrix	Create or link verification test evidence to each requirement and rescore readiness.
M4 retrieved evidence	11 candidate item(s) exist, but are not automatically controlled objective evidence.	M2 complaint pipeline and M4 evidence retrieval	Promote accepted snippets into controlled evidence nodes with requirement IDs, source, confidence, and firmware.
Trace Decay	2 firmware freshness alert(s) require review before release or approval.	M2 Trace Decay over M1 firmware/test links	Rerun verification on current firmware or document an approved equivalence rationale.
CAPA linkage	Open CAPA or complaint risk can keep readiness from being clean even when evidence exists.	M1 graph relationships and M2 risk assessment	Close, justify, or explicitly reference CAPA impact before final regulatory approval.

Reason and Corrective Action

The root issue is a traceability-linkage gap rather than a simple absence of all evidence. Corrective action is to synchronize M4 retrieved evidence back into the M1 graph as controlled verification evidence, then rerun AuditShadow and Trace Decay. Until that link exists, the report correctly treats the material as candidate support and keeps the audit finding open.

14.2 AuditShadow Detail Appendix

Finding(s)	Requirement(s)	Reference	Risk	Observation	Remediation
FINDING-020	REQ-SPO2-003	IEC 60601-1-8	Major	M1 graph has no VERIFIED_BY test evidence linked for this requirement. M4 may have retrieved candidate evidence, but it is not controlled objective verification evidence until Quality links it to the requirement, test record, source, and version/configuration metadata. [DEMO DATA - NOT CONTROLLED EVIDENCE]	Link or create verification test evidence with current version/configuration metadata, then rerun AuditShadow. [DEMO DATA - NOT CONTROLLED EVIDENCE]

Audit Interpretation

Major and critical findings should be triaged before report approval. A graph-linked evidence gap means the requirement is present in the M1 graph while still lacking a controlled verification test relationship acceptable for submission or inspection. Candidate RAG evidence must be reviewed, accepted, and linked before the finding can close.

14.3 Trace Decay Detail Appendix

Requirement(s)	Test(s)	Tested Firmware	Required Firmware	Reason
REQ-SPO2-003	TEST-003, TEST-023	v3.4	v3.4	Test is synthetic/demo backfill and cannot be treated as controlled verification evidence. [DEMO DATA - NOT CONTROLLED EVIDENCE]

Traceability Impact Statement

Trace Decay identifies verification evidence that may no longer represent the active or proposed firmware state. These alerts are not final nonconformities by themselves, but they are mandatory review inputs for release, CAPA, and audit-readiness decisions.

15. Cross-Standard Rendering Matrix

Element	FDA / QMSR	EU MDR	ISO 13485
Safety Requirement	Design input under QMSR; linked to verification evidence.	GSPR and Annex I safety/performance claim.	ISO 13485 Clause 7.3.3 design input.
Test Evidence	DHF verification record; current configuration/version required.	Annex II Section 6.1 V&V evidence.	Clause 7.3.6 design verification.
Complaint Record	Complaint file CPL-20260708-C257 with malfunction assessment.	Post-market vigilance signal under Articles 83/87.	Complaint handling under Clause 8.2.2.
CAPA Record	CAPA-20260708-DA21 opened for Alarm Notification, Measurement and Sensor Performance.	Trend/CAPA input for PMS and vigilance review.	Corrective action under Clause 8.5.2.
Risk Documentation	Risk analysis tied to design controls and complaint file.	Benefit-risk update in Annex II Section 5.	Risk management continuity through production/post-production.
Product Documentation	Lifecycle impact for Alarm Notification, Measurement and Sensor Performance.	Design information and change-control evidence.	Validation and change-control evidence required.

Regulatory Lens Note

The same live graph and agent state are rendered under three regulatory lenses. This report therefore does not duplicate content blindly; it reformats the same evidence chain for FDA/QMSR, EU MDR, and ISO 13485 review contexts.

16. Document Control and Approval

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Revision Date	2026-07-08
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CAPA ID	CAPA-20260708-DA21
Prepared by	DocPlus+ M4 Evidence and Documentation Layer
Generated from	Live M2 complaint pipeline, M1 graph context, M4 evidence retrieval, and device-level SBOM/NVD scan when attached
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Role	Name	Signature	Date
Quality Reviewer			
Regulatory Reviewer			